CHAPTER 21

TO THE SKY!

The stairs of the sky are let down for him that he may ascend thereon to heaven. O gods, put your arms under the king: raise him, lift him to the sky. To the sky! To the sky!

—HYMN FOR A DEAD PHARAOH (EGYPT, CA. 2600 B.C.)

When my grandparents were children, the electric light, the automobile, the airplane, and the radio were stupefying technological advances, the wonders of the age. You might hear wild stories about them, but you could not find a single exemplar in that little village in Austria-Hungary, near the banks of the river Bug. But in that same time, around the turn of the last century, there were two men who foresaw other, far more ambitious, inventions—Konstantin Tsiolkovsky, the theoretician, a nearly deaf schoolteacher in the obscure Russian town of Kaluga, and Robert Goddard, the engineer, a professor at an equally obscure American college in Massachusetts. They dreamt of using rockets to journey to the planets and the stars. Step by step, they worked out the fundamental physics and many of the details. Gradually, their machines took shape. Ultimately, their dream proved infectious.

In their time, the very idea was considered disreputable, or even a symptom of some obscure derangement. Goddard found that merely mentioning a voyage to other worlds subjected him to ridicule, and he dared not publish or even discuss in public his long-term vision of flights to the stars. As teenagers, both had epiphanal visions of